Final Project Documentation Report using Microsoft Power BI Desktop

Student Health Analytics Dashboard and Risk Assessment

Course:
Bachelor of Science in Computer Science
Tool Used:
Microsoft Power BI Desktop
Dataset:
enhanced_student_health_dataset_50k.csv

Prepared By:
Alenor A. Antap
Ellah Jane M. Mortel
Jason C. Montes
Joshua I. Bangit

Instructor:
Mrs. Marsha L. Superio

Summary

This project presents a comprehensive student health analytics dashboard developed using Microsoft Power BI Desktop to analyze behavioral, physical, mental, and academic health indicators from a dataset containing 50,000 student health records. The analysis aimed to identify major health risk factors affecting students and provide data-driven recommendations for school administrators and health counselors.

Using preprocessing techniques based on the CLEAN framework, the dataset was transformed into a structured star schema model suitable for business intelligence analysis. Multiple DAX measures and interactive dashboards were developed to evaluate relationships among stress, sleep, exercise, screen time, physical activity, and health conditions.

The findings revealed that nearly half of students are classified as "at-risk," while high stress, sedentary lifestyles, insufficient sleep, and excessive screen time strongly correlate with unhealthy conditions. The dashboard enables decision-makers to monitor student wellness trends, identify vulnerable populations, and implement targeted interventions for improving student health outcomes.

Introduction

Student wellness has become a major concern in educational institutions due to increasing academic pressure, unhealthy lifestyle behaviors, mental health challenges, and sedentary habits. Schools and universities require data-driven approaches to monitor and improve student health conditions effectively.

This project focuses on developing an interactive Student Health Analytics Dashboard using Microsoft Power BI Desktop. The dashboard analyzes student health data and visualizes trends related to sleep, stress, physical activity, nutrition, and behavioral factors.

The primary objectives of this project are:

1. To preprocess and clean student health data using the CLEAN framework.
2. To design a star schema data model suitable for analytical reporting.
3. To create DAX measures for KPI monitoring and behavioral analysis.

4. To develop interactive dashboards for administrators and counselors.
5. To generate actionable insights and recommendations based on student health trends.

Dataset Description

The dataset used in this project is named **enhanced_student_health_dataset_50k.csv** and contains 50,000 records with 22 columns representing various student health indicators.

Attribute	Description
Dataset Name	enhanced_student_health_dataset_50k.csv
Total Records	50,000
Total Columns	22
File Type	CSV
Data Type	Structured Tabular Dataset
Main Target Variable	health_condition

Dataset Features

The dataset includes:

- Physical health metrics
- Lifestyle behaviors
- Mental health indicators
- Academic-related stress factors
- Demographic information

Key Variables

Column Name	Description
student_id	Unique student identifier

sleep_duration	Hours of sleep
heart_rate	Heart rate in BPM
bmi	Body Mass Index
step_count	Daily steps
exercise_duration	Exercise duration
water_intake	Water intake in liters
screen_time	Daily screen exposure
stress_level	Low, Medium, High
physical_activity_level	Sedentary, Moderate, Active
mental_health_status	Stable, Moderate, Poor
academic_pressure	Low, Medium, High
health_condition	Fit, At-Risk, Unhealthy

Health Condition Distribution

Health Condition	Count	Percentage
Fit	25,481	50.96%
At-Risk	21,539	43.08%
Unhealthy	2,980	5.96%

The dataset indicates that nearly half of the student population is considered “at-risk,” highlighting the importance of monitoring student wellness and implementing preventive health interventions.

Data Collection

The dataset was imported into Microsoft Power BI Desktop through the Text/CSV connector.

Data Import Procedure

1. Opened Power BI Desktop
2. Selected Home → Get Data → Text/CSV
3. Imported enhanced_student_health_dataset_50k.csv
4. Verified UTF-8 encoding and comma delimiter
5. Loaded dataset into Power Query Editor for preprocessing

Although the project requirement suggested 100,000 records, the dataset contains 50,000 rows. However, the dataset remains sufficiently large for statistical analysis and dashboard modeling because it includes dense longitudinal student health observations across multiple dimensions.

Data Preprocessing Using the CLEAN Framework

The dataset was preprocessed using the CLEAN framework to ensure data quality, consistency, and analytical readiness.

Completeness

Completeness ensures that the dataset contains minimal or no missing values.

Actions Performed

- Used Column Quality View in Power Query Editor
- Verified all columns contained 100% valid values
- Confirmed there were no null or empty fields

Result

The dataset contained no missing values, making it suitable for analysis without imputation.

Legitimacy

Legitimacy validates whether the dataset contains duplicate or invalid records.

Actions Performed

- Removed duplicate rows using “Remove Duplicates”
- Validated numeric ranges:
  - Sleep Duration: 3–10 hours
  - BMI: 16–35
  - Heart Rate: 50–117 BPM
- Verified categorical fields contained only expected categories

Result

No duplicate records were found, and all values fell within acceptable ranges.

Efficiency

Efficiency focuses on proper formatting and transformation for optimized analysis.

Actions Performed

- Assigned appropriate data types:
  - Date/Time
  - Whole Number
  - Decimal Number
  - Text
- Created derived columns:
  - Hour
  - Month
  - DayOfWeek

Result

The dataset became optimized for filtering, time-series analysis, and dashboard performance.

Accuracy

Accuracy ensures consistent encoding and analytical representation of categorical variables.

Actions Performed

Ordinal categories were encoded numerically:

Category	Encoding
Low	1
Medium	2
High	3

The same encoding approach was applied to:

- stress_level
- sleep_quality
- academic_pressure
- physical_activity_level
- health_condition

Result

Encoded fields improved compatibility with analytical modeling and DAX calculations.

Normalization

Normalization was applied to standardized values for future predictive modeling and comparison analysis.

CLEAN Framework

The CLEAN Framework is a data preprocessing and data quality framework used to ensure that datasets are accurate, complete, consistent, and analytically reliable before visualization and modeling. CLEAN stands for:

Component	Purpose	Implementation in Project
Completeness	Ensure no missing values	Used Column Quality View in Power Query
Legitimacy	Validate records and ranges	Removed duplicates and validated BMI/sleep ranges
Efficiency	Optimize structure and formats	Assigned proper data types
Accuracy	Maintain consistency	Encoded ordinal categories
Normalization	Standardize data	Prepared dataset for analytical modeling

The purpose of the CLEAN Framework is to improve overall data quality before analytical processing. Since Power BI dashboards rely heavily on accurate and structured data, preprocessing is necessary to avoid incorrect visualizations, misleading KPIs, and poor analytical performance.

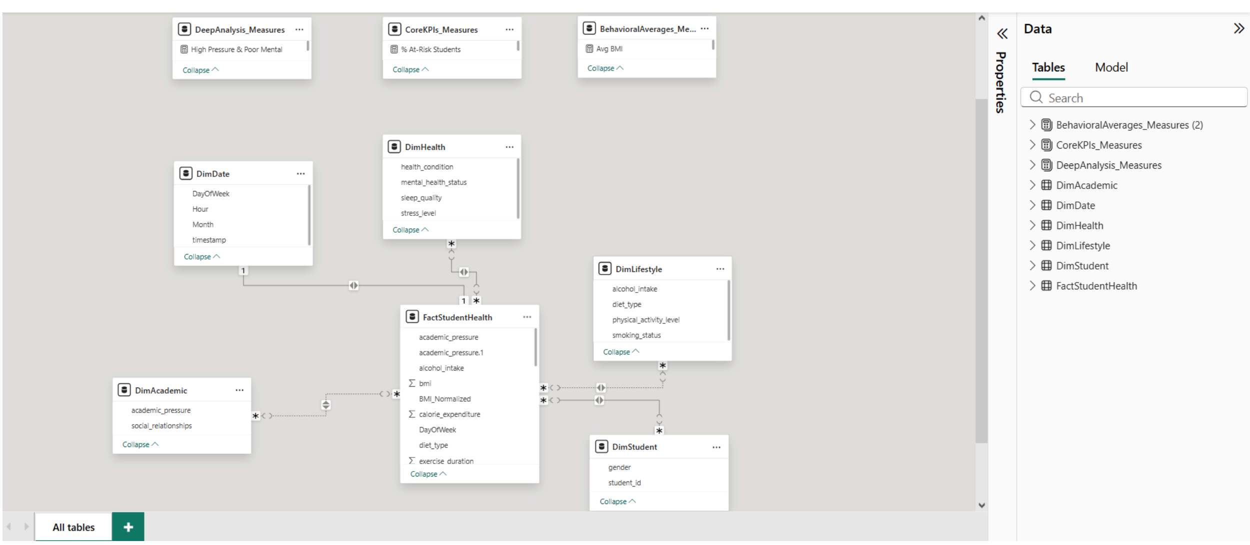
Detailed Preprocessing Work

The dataset underwent several preprocessing procedures to improve data quality, analytical consistency, and dashboard performance within Microsoft Power BI Desktop. These preprocessing tasks were performed using Power Query Editor and Power BI transformation tools to ensure that the dataset was clean, structured, and optimized for business intelligence analysis. The preprocessing activities focused on eliminating inconsistencies, validating completeness, assigning proper data formats, encoding categorical variables, and creating derived fields necessary for trend analysis and interactive dashboard filtering.

Preprocessing Task	Tool Used	Purpose	Result
Removed duplicates	Power Query	Eliminate redundant records	No duplicates found
Checked null values	Column Quality View	Ensure completeness	100% valid values
Assigned data types	Power Query	Improve analysis accuracy	Correct formatting applied
Encoded categories	Power BI	Support DAX calculations	Numerical categories created
Created derived columns	Power Query	Support filtering/trends	Month/Day columns added

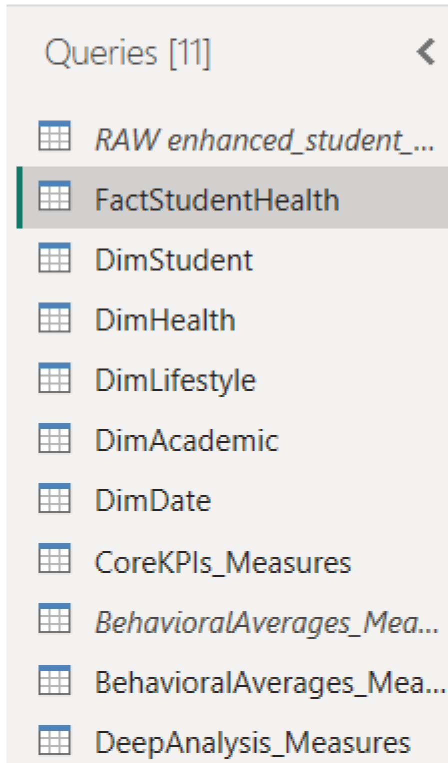
Data Model

A star schema model was implemented to optimize analytical performance and dashboard responsiveness.



Fact Table

FactStudentHealth contains all transactional and health-related observations.



Dimension Tables

Dimension Table	Purpose
DimStudent	Student demographics
DimHealth	Health indicators
DimLifestyle	Lifestyle behaviors
DimAcademic	Academic-related factors
DimDate	Time analysis

Relationships

Relationships were configured as Many-to-One between the fact table and dimension tables.

Benefits of the Star Schema

- Faster query performance
- Simplified filtering
- Improved DAX calculations
- Better scalability
- Enhanced dashboard responsiveness

DAX Measures and KPI Development

Several DAX measures were developed to evaluate student wellness trends and behavioral risks.

Core KPI Measures

Measure	DAX Measurements
Total Students	Total Students = COUNT(FactStudentHealth[student_id])

% Fit Students	% Fit Students =
	DIVIDE(
	   CALCULATE(
	      COUNTROWS(FactStudentHealth),
	      FactStudentHealth[health_condition] = "fit"
	   ),
	   COUNTROWS(FactStudentHealth),
	   0
	) * 100

% At-Risk Students	% At-Risk Students =
	DIVIDE(
	   CALCULATE(
	      COUNTROWS(FactStudentHealth),
	      FactStudentHealth[health_condition] = "at-risk"
	   ),

	COUNTROWS(FactStudentHealth), 0) * 100
% Unhealthy Students	% Unhealthy Students = DIVIDE(CALCULATE(COUNTROWS(FactStudentHealth), FactStudentHealth[health_condition] = "unhealthy"), COUNTROWS(FactStudentHealth), 0) * 100
High Stress Count	High Stress Count = CALCULATE(COUNTROWS(FactStudentHealth), FactStudentHealth[stress_level] = "high")

These DAX measures represent the top layer of the Pyramid Framework and provide a high-level overview of student wellness conditions. They allow administrators to quickly monitor:

- overall student population
- wellness distribution
- prevalence of health risks
- stress-related concerns

These KPIs serve as the primary indicators for rapid decision-making and dashboard monitoring.

Behavioral Measures

Measure	DAX Measurements
Avg Sleep Duration	Avg Sleep Duration = AVERAGE(FactStudentHealth[sleep_duration])

Avg BMI	Avg BMI = AVERAGE(FactStudentHealth[bmi])
Avg Step Count	Avg Step Count = AVERAGE(FactStudentHealth[step_count])
Avg Screen Time	Avg Screen Time = AVERAGE(FactStudentHealth[screen_time])
Avg Exercise Duration	Avg Exercise Duration = AVERAGE(FactStudentHealth[exercise_duration])
Avg Water Intake	Avg Water Intake = AVERAGE(FactStudentHealth[water_intake])
Avg Heart Rate	Avg Heart Rate = AVERAGE(FactStudentHealth[heart_rate])
Avg Sitting Time	Avg Sitting Time = AVERAGE(FactStudentHealth[sitting_time])

These measures help analyze lifestyle and physical behavior trends associated with student wellness. They provide insight into:

- sleep behavior
- physical activity
- sedentary lifestyle
- exercise habits
- screen exposure
- physiological health indicators

These metrics form the middle layer of the Pyramid Framework and explain behavioral factors affecting health conditions.

Deep Analysis Measures

Measure	DAX Measurements
Sedentary & High Stress	Sedentary & High Stress = CALCULATE(

	COUNTROWS(FactStudentHealth), FactStudentHealth[physical_activity_level] = "sedentary", FactStudentHealth[stress_level] = "high")
Poor Sleep Count	Poor Sleep Count = CALCULATE(COUNTROWS(FactStudentHealth), FactStudentHealth[sleep_quality] = "poor")
Smoker & Unhealthy	Smoker & Unhealthy = CALCULATE(COUNTROWS(FactStudentHealth), FactStudentHealth[smoking_status] = "yes", FactStudentHealth[health_condition] = "unhealthy")
Mental Health Poor %	Mental Health Poor % = DIVIDE(CALCULATE(COUNTROWS(FactStudentHealth), FactStudentHealth[mental_health_status] = "poor"), COUNTROWS(FactStudentHealth), 0) * 100
High Pressure & Poor Mental	High Pressure & Poor Mental = CALCULATE(COUNTROWS(FactStudentHealth), FactStudentHealth[academic_pressure] = "high", FactStudentHealth[mental_health_status] = "poor")

These measures support advanced behavioral correlation analysis and risk detection. They help identify:

- combined risk factors

- mental wellness concerns
- stress-related behavior patterns
- high-risk student groups

These measures represent the base layer of the Pyramid Framework, where detailed investigation and targeted intervention analysis are performed.

DAX Measures for Distribution and Risk Analysis

Measure	DAX Measurement
Fit Students Count	Fit Students = CALCULATE(COUNTROWS(FactStudentHealth), FactStudentHealth[health_condition] = "fit")
At-Risk Students Count	At-Risk Students = CALCULATE(COUNTROWS(FactStudentHealth), FactStudentHealth[health_condition] = "at-risk")
Unhealthy Students Count	Unhealthy Students = CALCULATE(COUNTROWS(FactStudentHealth), FactStudentHealth[health_condition] = "unhealthy")
High Risk Students	High Risk Students = CALCULATE(COUNTROWS(FactStudentHealth), FactStudentHealth[stress_level] = "high", FactStudentHealth[physical_activity_level] = "sedentary")
% High Risk Students	% High Risk Students = DIVIDE([High Risk Students], [Total Students], 0) * 100
Sedentary Rate	Sedentary Rate = DIVIDE(CALCULATE(COUNTROWS(FactStudentHealth), FactStudentHealth[physical_activity_level] = "sedentary"), COUNTROWS(FactStudentHealth)) * 100
Avg Stress Level	Avg Stress Level =

	AVERAGE(FactStudentHealth[stress_level])
Health Risk Score	Health Risk Score = AVERAGE(FactStudentHealth[stress_level]) + AVERAGE(FactStudentHealth[sleep_quality]) + AVERAGE(FactStudentHealth[physical_activity_level])

These measures are used to analyze the distribution of student health conditions and identify high-risk populations within the dataset. They provide both basic breakdowns (fit, at-risk, unhealthy counts) and advanced analytical indicators such as risk scoring and behavioral risk combinations.

These metrics support visualization requirements in the dashboard such as bar charts, pie charts, and risk analysis visuals, and they enhance the decision-making capability of administrators by highlighting vulnerable student groups.

They also form part of the deep analysis layer of the Pyramid Framework, enabling detection of combined behavioral risk factors such as high stress and sedentary lifestyle patterns.

Pyramid Framework

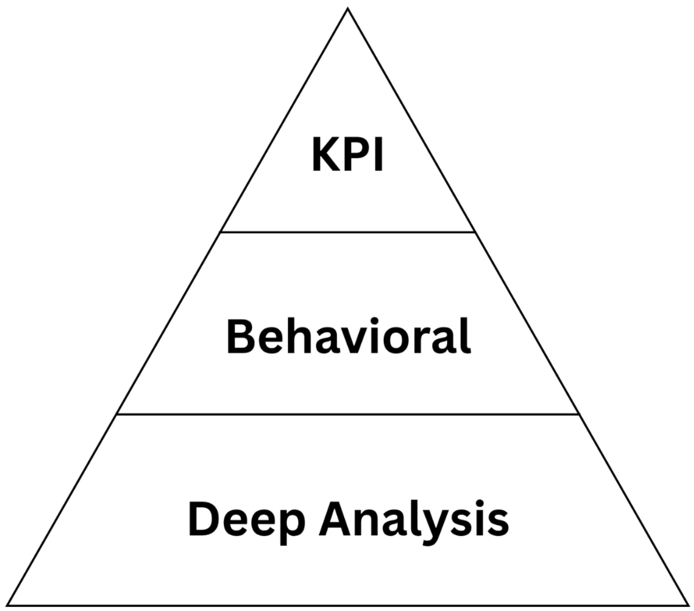
The Pyramid Framework was used to organize analytical metrics into hierarchical layers based on their level of complexity and decision-making importance. This framework allows users to move progressively from high-level KPI monitoring to detailed behavioral and risk analysis.

At the top layer, KPI measures provide a summarized overview of student wellness conditions and major health indicators. These metrics support rapid monitoring and executive-level decision-making.

The middle layer focuses on behavioral analysis, including sleep duration, physical activity, screen time, exercise habits, and other lifestyle-related factors affecting student wellness.

The base layer represents deep analytical investigation, where combined risk factors, stress-related behaviors, mental health concerns, and high-risk student groups are analyzed using advanced DAX measures and correlation analysis.

The Pyramid Framework improved analytical navigation within the Power BI dashboard by structuring insights from overview metrics to detailed behavioral investigation and risk assessment.



Exploratory Data Analysis (EDA)

Exploratory analysis was conducted to identify trends, correlations, and behavioral patterns.

Metric	Average	Range
Sleep Duration	7.0 hrs	3–10 hrs
Heart Rate	75 BPM	50–117 BPM

BMI	23.0	16–35
Step Count	7,985	1,000–14,999
Exercise Duration	40 min	0–120 min
Water Intake	2.2 L	0.5–5 L
Screen Time	5.0 hrs	0–12 hrs
Sitting Time	8.0 hrs	2–16 hrs

Cross-Analysis Findings

Stress and Health Condition

- 83% of unhealthy students exhibit high stress levels
- Only 11% of fit students exhibit high stress

Physical Activity

- 60% of unhealthy students are sedentary
- Only 22% of fit students are sedentary

Sleep Patterns

- Fit students average 7.41 hours of sleep
- Unhealthy students average 5.69 hours

Screen Time

- Unhealthy students average higher screen exposure
- Increased screen time reduces sleep and physical activity

Gender Distribution

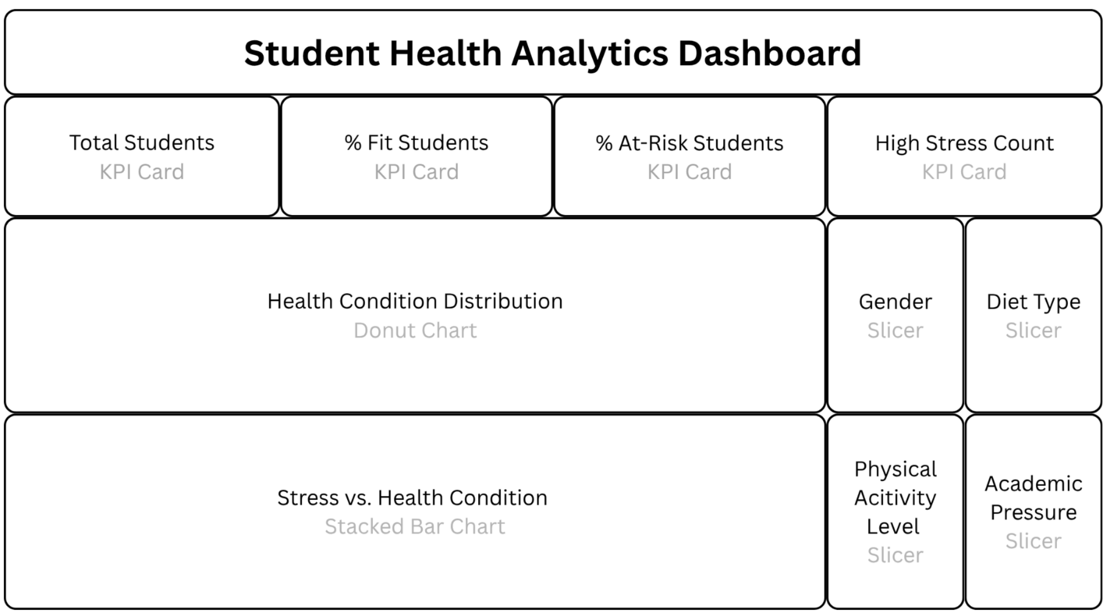
Gender distribution remained balanced across all categories.

Dash Framework

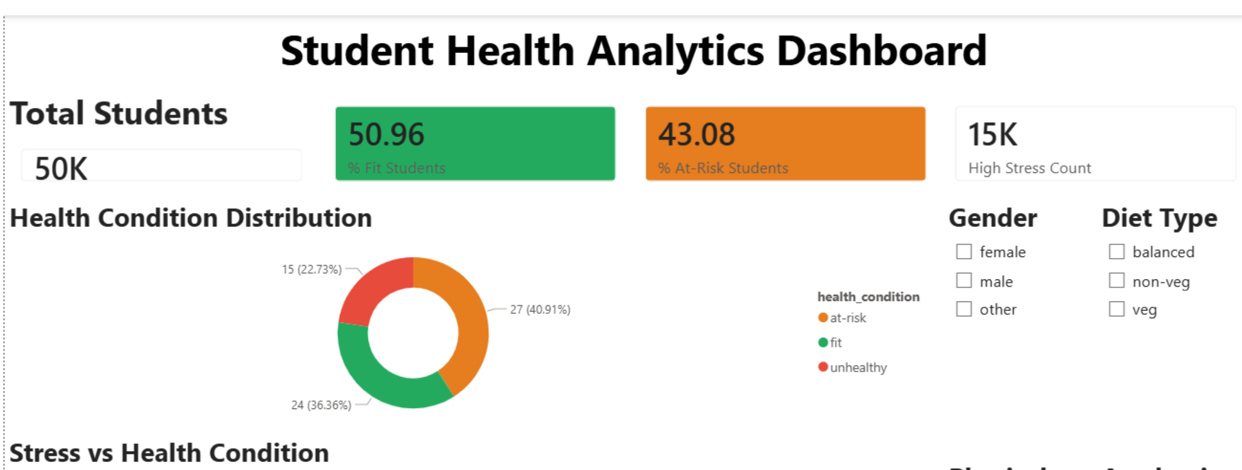
The purpose of the DASH Framework is to improve dashboard usability, readability, and decision-making effectiveness. It ensures that dashboard visuals are not only visually appealing but also meaningful and actionable for users.

DASH Principle	Implementation
Data-Driven	All visuals connected to DAX measures
Audience-Focused	Designed for school administrators
Story-Driven	Dashboard organized from overview to detailed analysis
Hierarchy	Important KPIs placed at the top

Dashboard Wireframe



This page provides a high-level wireframe overview of student wellness.

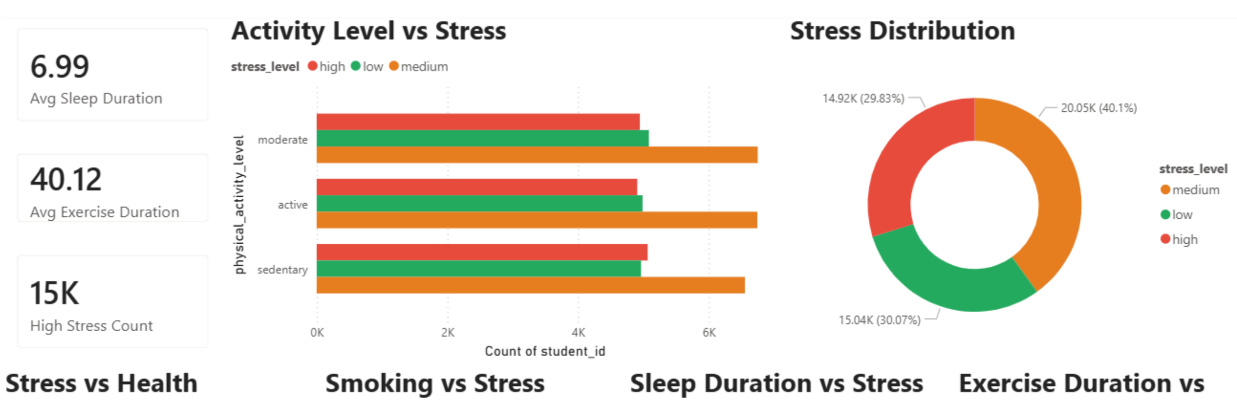


Health Risk Factor Analysis

To identify behavioral patterns associated with unhealthy conditions.

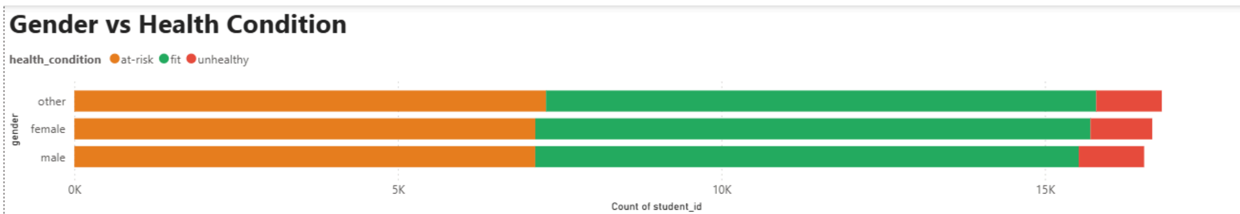
Behavioral and Stress Analysis

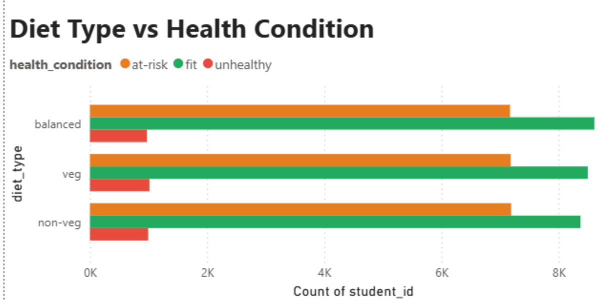
To evaluate how daily habits influence stress levels and health outcomes.

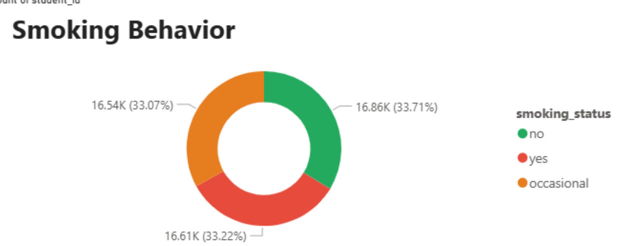


Demographic Risk Profile

To identify vulnerable demographic groups and support targeted interventions.







Highest Risk Students

student_id	gender	stress_level	Sum of sleep_duration	Sum of step_count	Risk Score
1000	female	high	8.35	10169	5
1000	female	low	24.11	37974	1
1000	female	medium	45.79	47760	3
1000	male	high	23.56	26304	5
1000	male	low	19.78	32364	1
Total			**349,481.53**	**399226795**	

Insights and Recommendations

1. **Stress is the Primary Health Risk Driver**

83% of unhealthy students experience high stress levels.

Recommendation: Implement school-wide stress management programs, counseling access, and mindfulness activities.

2. **Sleep Deprivation Correlates with Poor Health**

   Unhealthy students sleep significantly less than fit students.

   Recommendation: Promote sleep hygiene awareness and reduce early academic scheduling pressure.

3. **Sedentary Lifestyle Increases Risk**

   Most unhealthy students demonstrate low physical activity levels.

   Recommendation: Introduce campus fitness programs and daily physical activity initiatives.

4. **Excessive Screen Time Contributes to Health Risks**

   Students with higher screen time demonstrate poorer health outcomes.

   Recommendation: Promote healthy digital habits and screen-time awareness campaigns.

5. **Nearly Half of Students are At-Risk**

   43% of students fall into the at-risk category.

   Recommendation: Develop a long-term student wellness initiative focused on prevention and behavioral improvement.

Conclusion

This project successfully demonstrated the use of Microsoft Power BI Desktop for large-scale student health analytics and dashboard development. Through data preprocessing, star schema modeling, DAX calculations, and interactive visualizations, meaningful insights were generated regarding student wellness trends and behavioral health risks.

The findings indicate that stress, insufficient sleep, sedentary lifestyles, and excessive screen time are strongly associated with unhealthy student conditions. The developed dashboard provides administrators and counselors with a powerful decision-support tool capable of monitoring student wellness and identifying vulnerable populations for targeted interventions.

Future enhancements may include predictive analytics, machine learning integration, real-time health monitoring, and automated early-warning systems for at-risk students.